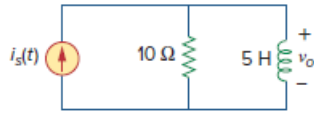


Problem

Find v_o for the circuit in Fig. 10.86, assuming that $i_s(t) = 2 \sin(2t) + 3 \cos(4t)$ A.

Figure 10.86:



Step-by-step solution

Step 1 of 7

Refer to Figure 10.86 in the textbook.

From Figure 10.86 in the textbook, the circuit operates at two different frequencies.

One way to obtain a solution is to use superposition, which breaks the problem into single – frequency problems.

So let,

$$v_o = v_1 + v_2 \dots\dots (1)$$

Where,

v_1 is due to the $2 \sin(2t)$ A current source,

v_2 is due to the $3 \cos(4t)$ A current source.

Step 2 of 7

Determine the value of voltage, v_1 .

Consider the source is, $i_s = 2 \sin(2t)$.

Convert the source into cosine form.

$$\begin{aligned} i_s &= 2 \sin(2t) \\ &= 2 \cos(2t - 90^\circ) \text{ A} \end{aligned}$$

Determine the value of inductive reactance, X_L .

$$\begin{aligned} X_L &= j\omega L \\ &= j(2)(5) \\ &= j10 \Omega \end{aligned}$$

Re-draw the circuit diagram.

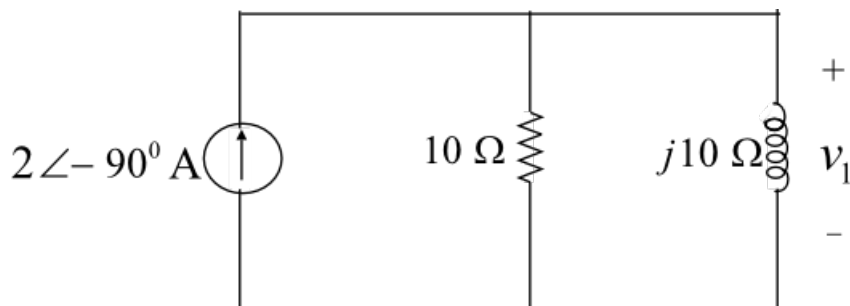


Figure 1

Step 3 of 7

Apply Kirchhoff's current law at node v_1 .

$$2\angle -90^\circ = \frac{v_1}{10} + \frac{v_1}{j10}$$

$$2\angle -90^\circ = v_1(0.1 - j0.1)$$

$$\begin{aligned} v_1 &= \frac{20\angle -90^\circ}{1 - j} \\ &= \frac{20\angle -90^\circ}{1.414\angle -45^\circ} \\ &= 14.14\angle -45^\circ \text{ V} \end{aligned}$$

Convert the equation to time-domain form.

$$\begin{aligned} v_1 &= 14.14 \cos(2t - 45^\circ) \\ &= 14.14 \sin(2t - 45^\circ + 90^\circ) \end{aligned}$$

$$v_1 = 14.14 \sin(2t + 45^\circ) \text{ V} \dots\dots (2)$$

Step 4 of 7

Determine the value of voltage, v_2 .

Consider the source is, $i_s = 3 \cos(4t)$.

Determine the value of inductive reactance, X_L .

$$\begin{aligned} X_L &= j\omega L \\ &= j(4)(5) \\ &= j20 \Omega \end{aligned}$$

Re-draw the circuit diagram:

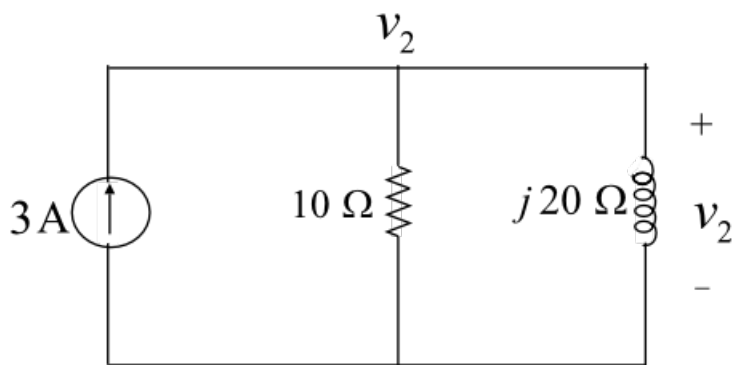


Figure 2

Step 5 of 7

Apply Kirchhoff's current law at node v_2 .

$$3 = \frac{v_2}{10} + \frac{v_2}{j20}$$

$$3 = 0.1v_2 - j0.05v_2$$

Step 6 of 7

Simplify the equation further,

$$\begin{aligned}v_2 &= \frac{3}{0.1 - j0.05} \\&= 24 + j12 \\&= 26.83 \angle 26.57^\circ \text{ V}\end{aligned}$$

Convert the equation to time-domain form.

$$v_2 = 26.83 \cos(4t + 26.57^\circ) \text{ V} \dots\dots (3)$$

Step 7 of 7

Substitute equation (2) and equation (3) in equation (1).

$$v_0 = 14.14 \sin(2t + 45^\circ) + 26.83 \cos(4t + 26.57^\circ) \text{ V}$$

Therefore, the value of voltage v_0 is,

$$\boxed{14.14 \sin(2t + 45^\circ) + 26.83 \cos(4t + 26.57^\circ) \text{ V}}.$$